

Brainstorm & Idea Prioritization

Ideation Phase

Date	07 March 2026
Team ID	
Project Name	Insurance Fraud Detection Using Machine Learning
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Step 1: Problem Statement Selected

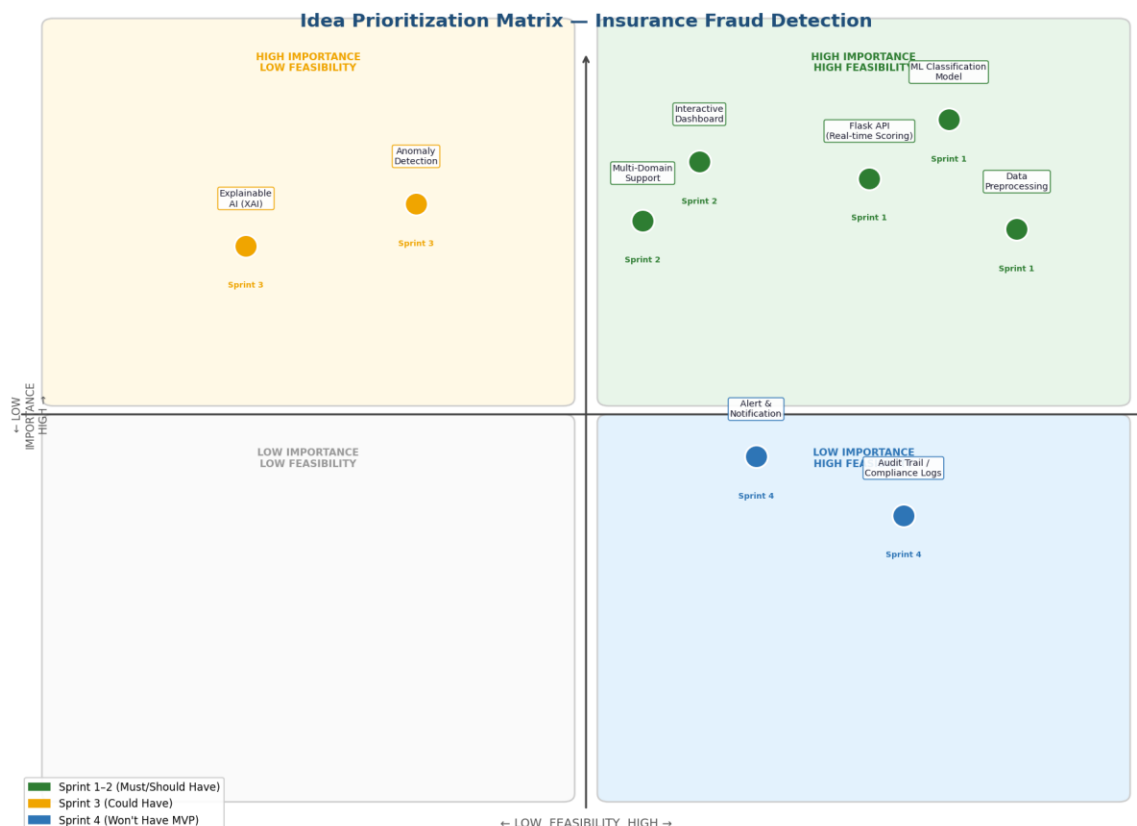
Field	Details
Problem Statement	How might we build an intelligent system that automatically detects fraudulent insurance claims across automobile, health, and property domains with high accuracy and minimal manual intervention?
Target Users	Insurance Claims Analysts, Fraud Reviewers, Risk Managers
Session Format	Team brainstorming (2–8 people), 1 hour collaboration
Key Rules	Stay on topic · Defer judgment · Go for volume · Encourage wild ideas · If possible, be visual

Step 2: Brainstormed Ideas

#	Idea	Category	Description
1	ML Classification Model	Core Feature	Train supervised ML models (Random Forest, XGBoost, Logistic Regression) on historical labeled claim data to classify each claim as Fraud or Genuine with a probability score.
2	Real-time Claim Scoring via API	Core Feature	Build a Flask REST API endpoint that receives claim data, passes it through the per-trained ML pipeline, and returns a fraud probability score in under 2 seconds.
3	Data Preprocessing Pipeline	Core Feature	Automate feature engineering, categorical encoding, outlier removal, and normalization of raw claim data before feeding it to the ML model.
4	Interactive Fraud Dashboard	UI/Visualization	Create a web-based dashboard using Chart.js/Plotly that shows real-time fraud trend graphs, claim volumes by category, and model performance metrics (accuracy, precision, recall).
5	Multi-Domain Model Support	Scalability	Develop separate, domain-specific ML models for automobile, health, and property insurance, each trained on domain-relevant features for higher accuracy.
6	Anomaly Detection Module	Advanced ML	Use unsupervised ML (Isolation Forest, Autoencoder) to detect unusual claim patterns that fall outside normal distributions — catching novel fraud types not in training data.

#	Idea	Category	Description
7	Explainable AI (XAI)	Trust & Transparency	Integrate SHAP or LIME to show claims officers which features (e.g., claim amount, claim frequency, time since policy start) most influenced the fraud prediction.
8	Fraud Alert & Notification System	Operations	Automatically send email/in-app alerts to assigned claims officers whenever a claim receives a fraud probability score above a configurable threshold (e.g., 0.8).
9	Historical Audit Trail	Compliance	Log all prediction events (input features, model version, score, reviewer decision) to a tamper-proof database for regulatory compliance and model monitoring.
10	Feedback Loop for Model Retraining	AI/ML Ops	Allow claims officers to flag incorrect predictions; accumulate this feedback to periodically retrain the ML model with newly labelled data for continuous improvement.
11	Mobile-Responsive Web Interface	Accessibility	Ensure the claim submission and review UI is fully responsive for mobile/tablet so officers can review flagged claims on the go.
12	Batch Claim Processing	Performance	Support bulk CSV upload of claims for offline batch fraud scoring, useful for processing large backlogs during system migration or audits.

Step 3: Idea Prioritization Matrix



Idea	Importance	Feasibility	Priority	Target Sprint
ML Classification Model	High	High	Must Have	Sprint 1
Real-time Claim Scoring via API (Flask)	High	High	Must Have	Sprint 1
Data Preprocessing Pipeline	High	High	Must Have	Sprint 1
Interactive Fraud Dashboard	High	Medium	Should Have	Sprint 2
Multi-Domain Model Support	High	Medium	Should Have	Sprint 2
Anomaly Detection Module	Medium	Medium	Could Have	Sprint 3
Explainable AI (XAI — SHAP/LIME)	Medium	Low	Could Have	Sprint 3
Fraud Alert & Notification System	Medium	Medium	Could Have	Sprint 4
Historical Audit Trail	Low	High	Won't Have (MVP)	Sprint 4
Feedback Loop / Model Retraining	Low	Low	Won't Have (MVP)	Future
Mobile-Responsive UI	Medium	High	Should Have	Sprint 2
Batch Claim Processing	Low	Medium	Won't Have (MVP)	Future